

HDOC Day-24

Write a C program that displays the following menu and uses a switch statement to print the “Increasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

Note: For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.

1. Increasing right-angled triangle

Input the number of rows, n, and print an increasing right-angled triangle using asterisks.

For n = 4:

```
*
* *
* * *
* * * *
```

2. Increasing row-number triangle

Input the number of rows, n, and print a triangle in which each row number is printed as many times as its value.

For n = 4:

```
1
2 2
3 3 3
4 4 4 4
```

3. Increasing natural-number pattern

Input the number of rows, n, and print natural numbers from 1 to the corresponding row number in each row.

For n = 4:

```
1
1 2
1 2 3
1 2 3 4
```

4. Consecutive-number triangle

Input the number of rows, n, and print consecutive natural numbers in a triangular pattern.

For n = 4:

```
1
2 3
4 5 6
7 8 9 10
```

5. Increasing alphabet pattern

Input the number of rows, n, and print uppercase English letters from A up to the letter corresponding to each row. Assume that $1 \leq n \leq 26$ for the alphabet pattern.

For n = 4:

```
A
A B
A B C
A B C D
```